

PRIYANSH TIWARI

IoT Implementation Engineer | Full-Stack Developer

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Summary

Recent ECE graduate specializing in Industrial IoT, embedded systems, and full-stack development. Experienced in building machine-monitoring and data-acquisition platforms using React.js, FastAPI, Node.js, ESP32, Raspberry Pi, and Vicharak Axon. Skilled at bridging hardware and software to deliver scalable, production-ready IoT applications. **Portfolio**

Education

Bachelor of Technology in Electronics and Communication Engineering

Gyan Ganga College of Technology

2022 – 2026

Jabalpur, Madhya Pradesh

- CGPA: 8.0/10

Higher Secondary Education (Class XII)

Vivekanand International Public School

2020 – 2022

Jabalpur, Madhya Pradesh

- Score: 78%

Experience

ManaraLabs

IoT Implementation Engineer Intern

July 2026 – Present

Indore, Madhya Pradesh

- Design IIoT solutions by integrating **ESP32** and **Vicharak Axon** microcontrollers, enabling continuous data acquisition across five production machines.
- Build responsive dashboards in **React.js** with backend services powered by **Fastify**, **FastAPI**, **Express.js**, and **Node.js** to process and visualize live industrial data, cutting manual reporting time by 20%.
- Implement **MQTT**, REST APIs, and WebSockets for reliable device communication, powering ongoing production tracking and machine-status alerts that trim fault-detection time by 15%.

STEM Learning

STEM Education Apprentice

October 2025 – December 2025

Seoni, Madhya Pradesh

- Delivered hands-on training in Arduino, embedded systems, robotics, and sensor interfacing to 50+ students and educators.
- Facilitated workshops on microcontroller programming and project-based STEM learning, raising student participation by 25%.
- Created practical learning activities that strengthened problem-solving skills, improving hands-on skill retention by 30% among first-time learners.

Projects

AI-Powered Smart Waste Segregation and Monitoring System | Github

NVIDIA Jetson Nano, Python, PyTorch, OpenCV, ESP32, IoT

August 2025 – September 2025

- Designed an AI-powered waste segregation system on **NVIDIA Jetson Nano** to classify biodegradable, recyclable, and non-recyclable waste through real-time computer vision, achieving 90% classification accuracy.
- Engineered a **PyTorch** and **OpenCV** image-classification pipeline and integrated **ESP32** sensors for automated bin-status monitoring, reducing manual sorting effort by 30%.
- Co-authored this work as a peer-reviewed book chapter, “**Design and Implementation of an AI-Enabled Waste Segregation and Monitoring System**,” published in *Advances in Electronics and Communication Systems: Design, Applications, and Emerging Technologies* (Volume 6, 2026).

Insightvillee – AI-Powered IIoT Platform | Backend & AI Systems

React, Node.js, Express, MongoDB, Mastra, OpenAI, WebSocket

August 2026

- Engineered a multi-turn **AI requirements-to-architecture pipeline** using **Mastra** and **OpenAI**, enabling conversational clarification before generating traceable solution proposals with cost-optimized bill-of-materials.
- Built a **Modbus RS-485 register discovery engine** that decodes IEEE 754 floating-point telemetry across four byte orderings, autonomously classifying sensor readings with high-confidence pattern recognition.
- Designed a secure, agentic diagnostic terminal enforcing a strict command allowlist and zero subprocess execution, validated through adversarial testing against a live language model with 100% attack rejection.

Technical Skills

- **Languages:** Java, Python, JavaScript, TypeScript, C++, Embedded C
- **Web & Backend Development:** React.js, HTML5, CSS3, Tailwind CSS, Vite, Node.js, Express.js, FastAPI, REST APIs, WebSockets, Accessibility (WCAG)
- **Databases:** MongoDB, MySQL, PostgreSQL
- **IoT, Embedded & Hardware:** ESP32, ESP8266, Arduino Uno, Arduino Nano, Arduino Nano 33 BLE Sense (TinyML), STM32, Raspberry Pi, Vicharak Axon, NVIDIA Jetson Nano
- **Protocols:** MQTT, HTTP/HTTPS, WebSockets, ModbusTCP, SSH, I2C
- **Tools & Platforms:** Git, GitHub, Docker, AWS, Linux, Arduino IDE, PlatformIO, Edge Impulse, VS Code, Postman

Achievements & Certifications

- Recognized as co-author of a **published book** chapter on **AI-powered waste segregation research** (Volume 6, 2026).
- Completed hands-on certification training in **5G Communication Technology**, conducted by **BSNL** in Jabalpur.
- Won the **Best Project Award** at **National Science Day** for an AI-based and IoT-based project.